



Supplemental Materials

for

An Introduction to CRISPR-Mediated Genome Editing in Fungi

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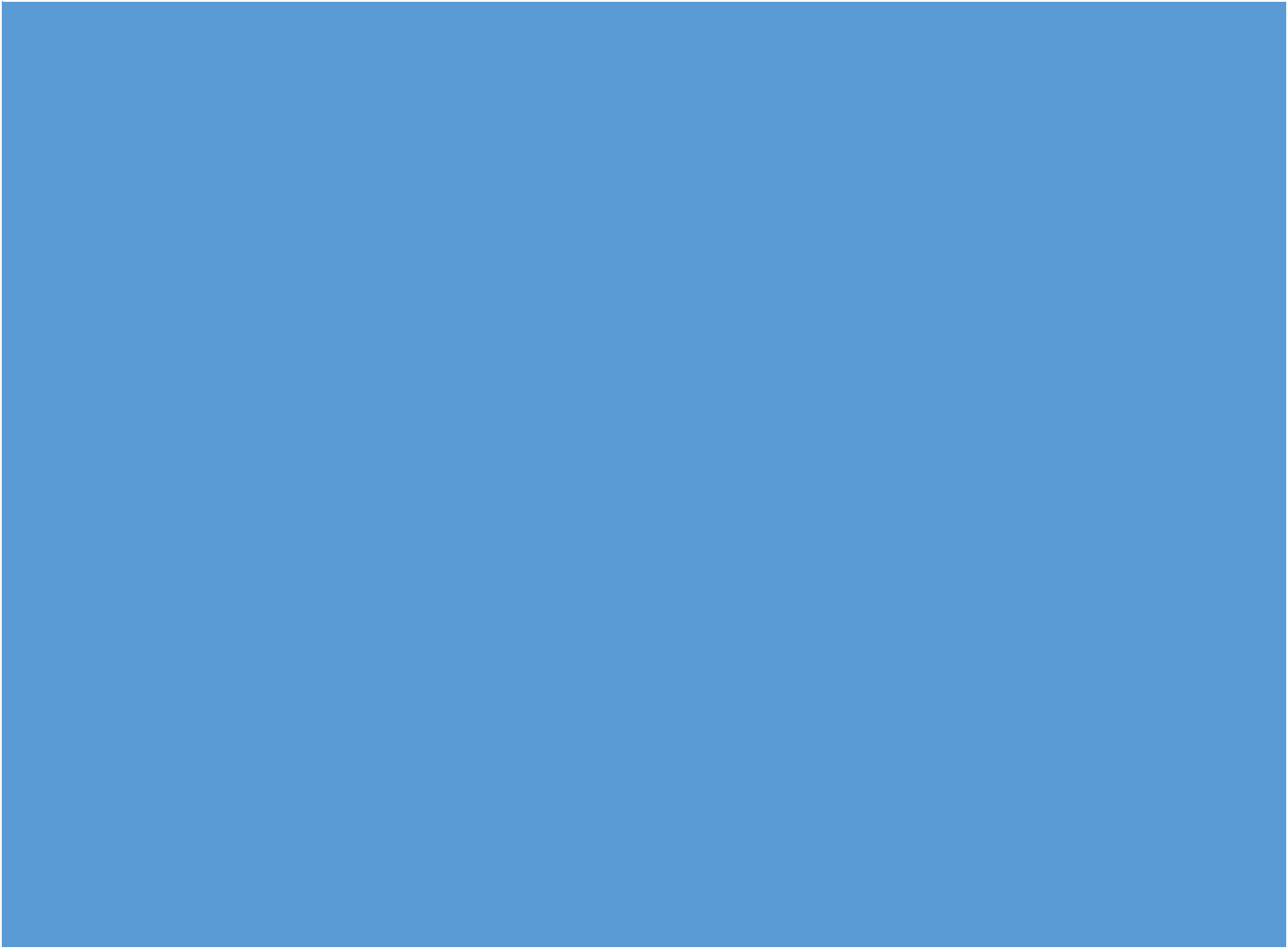
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An introduction to CRISPR-mediated genome editing in fungi

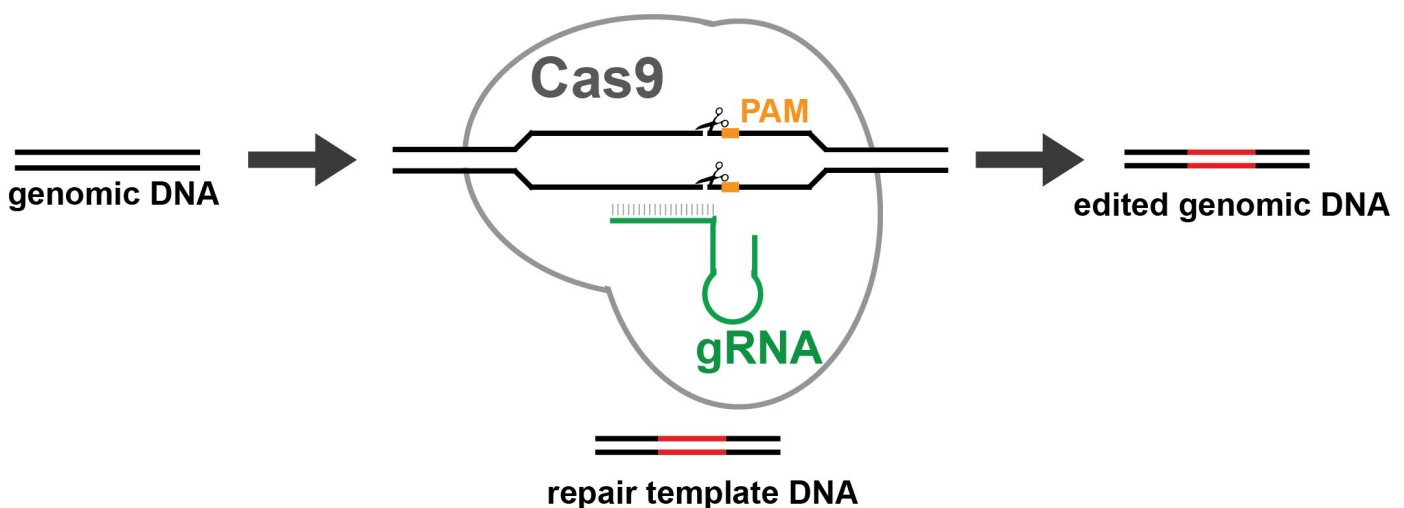


Appendix 1: Laboratory Exercises

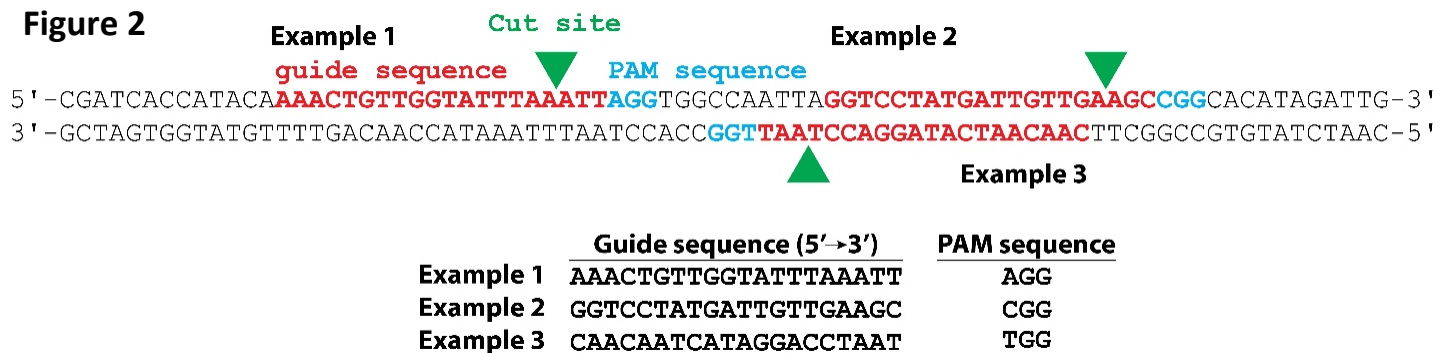
Introduction: Genetic modification or mutagenesis of organisms is one tool scientists use to investigate organism phenotype. Mutagenesis can be separated into two broad categories: random and targeted. In random mutagenesis, a chemical or energy source damages DNA randomly throughout the genome. As the damage is repaired, permanent changes to the DNA sequence may be introduced. In targeted mutagenesis, damage is directed to a specific DNA sequence thus changes made during repair will be isolated to the targeted DNA sequence.

One way to perform targeted mutagenesis is genome editing using Cas9, which is part of the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) system. In bacteria, CRISPR systems form a type of immune system, which defends the host by cleaving invading viral genetic material. Scientists are leveraging the power of CRISPR systems to edit the genomes of a wide variety of organisms. Application of this technology is fundamentally changing both industrial and academic science. To edit a genome with CRISPR an investigator needs 3 things: **1)** a nuclease that will cut the genome, most commonly Cas9 **2)** A 20 base guide RNA (gRNA), that will guide Cas9 to the specific genome sequence using base pairing **3)** A repair template in coordination with the host DNA repair machinery directs the repair of the double-strand break introduced by Cas9, and in doing so introduces a mutation (edit) to the genome (**Figure 1**).

Figure 1



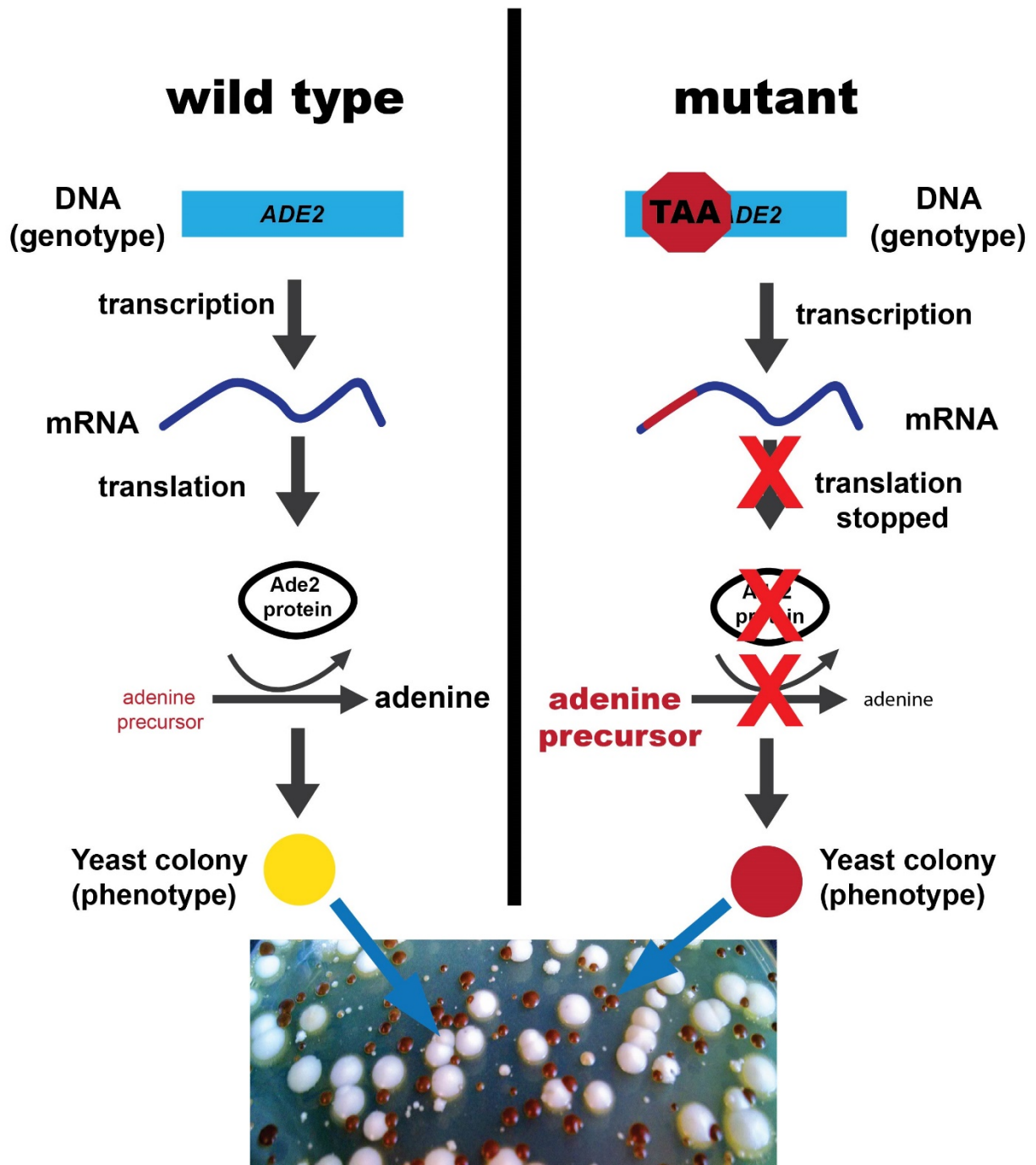
The precision of this technique comes from the specificity of the 20 base guide RNA, which binds to complementary DNA bases at the target site. Since there are 4 possible bases in DNA (represented as A, G, C, and T), there are 4^{20} different 20 base pair sequence combinations— that’s over a trillion! The human genome is between 3-4 billion bases and the yeast *Saccharomyces cerevisiae* genome is roughly 12 million bases long. A 20 base pair sequence is enough to uniquely target a large fraction of the genome, while repeated sequences can be more challenging. It is easy and inexpensive to change the sequence CRISPR targets by changing the sequence of the guide RNA. In addition, for cutting to happen at any given 20 base pair sequence, there is one additional requirement – the presence of a 3’ protospacer adjacent motif (PAM) sequence (**Figure 2**). For Cas9, the PAM sequence is NGG, where N is any base (A, G, C, or T). In the examples below (**Figure 2**), three

Figure 2

distinct PAM sequences are labeled. PAM sequences offer an additional level of protection against nonspecific cleavage, since DNA cleavage will not happen if they are not present. After the Cas9/gRNA complex finds its corresponding DNA binding site, cleavage occurs on both DNA strands. Green triangles indicate cleavage sites. The cell’s DNA repair machinery uses the repair template to fix the double stranded break. The repair template must disrupt the Cas9/gRNA nuclease recognition elements (gRNA binding site and/or PAM) for genome editing to occur. If the recognition elements were to remain, the nuclease could potentially bind and cleave the target sequence again.

In this lab we are going to edit the genome of the budding yeast *S. cerevisiae*. People have been using yeast for thousands of years for making bread, wine, and chocolate – in recent years, it has also proven useful for the production of fuel ethanol, medicines (e.g., insulin), as well as important commodity chemicals from renewable sources (instead of fossil fuels). Much of our current understanding of basic cellular processes including DNA replication, cell cycle, and transcription were first established in yeast, and to this day it remains an important model organism for processes that are conserved through eukaryotes, including humans. As a model, *S. cerevisiae* has several important characteristics: it grows quickly, has a well described genome, displays readily observable phenotypes, and is inexpensive to maintain. We will use CRISPR to insert a stop codon into the *ADE2* gene (part of the adenine biosynthesis pathway) and examine how mutagenesis of *ADE2* changes *S. cerevisiae* phenotype. Mutations in *ADE2* lead to the buildup of a precursor in the adenine biosynthesis pathway that turns the yeast a bright red color (**Figure 3**). (It should be noted that while this exercise targets *ADE2*, any gene can be targeted by CRISPR in *S. cerevisiae* or a number of other yeast species by altering the guide and repair template sequences.)

Figure 3



Objectives of the Investigation

Upon completion of this investigation, you should be able to:

List and Describe roles for each component of CRISPR mediated genome editing.

Identify PAM sequences and Guide RNA (gRNA) sequences in DNA.

Outline the steps of a yeast transformation.

Perform CRISPR mediated genome editing on *S. cerevisiae*.

Describe how changes in genotype caused by CRISPR effect *S. cerevisiae* phenotype.

Analyze *S. cerevisiae* transformants to determine which have been edited by CRISPR.

Compare and Contrast Nonhomologous End Joining and Homologous Recombination as mechanisms of DNA cleavage repair.

Discuss how genome editing could be used to improve human life and the challenges the scientific community faces to ensure genome editing is used responsibly.

Pre Lab Questions:

1. What organism will we be mutating in this lab?
2. What gene will we be mutating in this lab, and how will we be mutating it?
3. How will we distinguish organisms that have a mutation in the target gene vs those who do not?
4. In the sequence in Figure 2, find five additional guide+PAM sequences that are not already labeled:

<u>Guide Sequence (5'→3')</u>	<u>PAM sequence</u>
1.	
2.	
3.	
4.	
5.	

5. Why must the repair template destroy the recognition sequence for genome editing to occur? What are the two types of recognition sequences that can be changed to accomplish this?
6. Find and read an article in the news that refers to CRISPR. Be prepared to discuss during the incubation and heat shock portions of the laboratory.

Materials:

- 1.5 ml tubes
- Micropipette
- Micropipette tips
- heat block or water bath at 42°C
- spectrophotometer
- glass beads or cell spreader
- *S. cerevisiae* strain BY4741 or supermarket/home brewing yeast
- Synthetic Complete –Uracil (SC -Ura) media for strain BY4741, or YPD+Nat (nourseothricin) media for other yeast strains
- **PLATE Buffer**
40% PEG 3350
100mM Lithium Acetate pH 7.5
10mM Tris-Cl pH 7.5
1mM EDTA pH 7.5
- **TE/LiOAc Buffer**
10mM Tris-Cl pH 7.5
1mM EDTA pH 7.5
100mM Lithium Acetate pH 7.5
- **Salmon Sperm DNA**
- **Transforming DNA** plasmids pV1382 and pVG1 (**Figure 4**) greater than 10µg is best (**For plasmid key see below**)

Plasmid Key:

CEN/ARS – yeast centromere sequence enables replication of the plasmid in *S. cerevisiae*

URA3 - auxotrophic marker, allows BY4741 which normally cannot grow without uracil to grow without uracil

CaCas9 – gene encoding the Cas9 nuclease

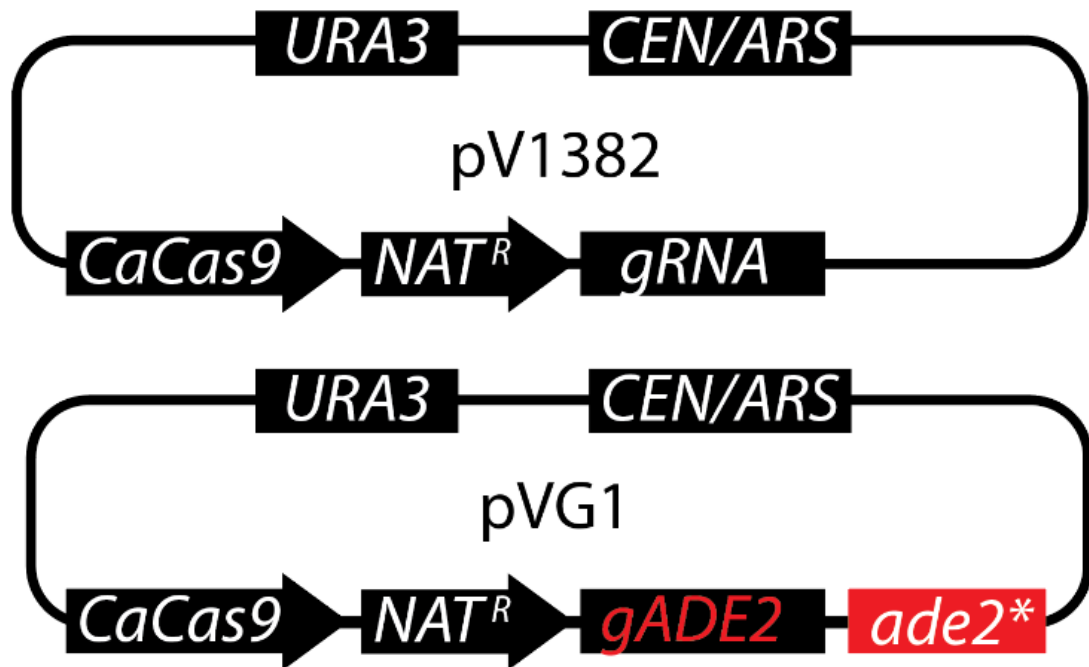
NAT^r - antibiotic resistance marker, allows for yeast to grow in the presence of the antibiotic nourseothricin

gRNA – site where a guide RNA could be cloned

gADE2 – guide RNA targeting *ADE2*

ade2* – repair template that places a stop codon in *ADE2*

Figure 4



Methods:

***S. cerevisiae* Transformation Protocol:**

1) Grow overnight YPD culture of *S. cerevisiae* at room temperature, ideally to Optical Density 600nm (OD₆₀₀) less than 6. Serial dilutions of overnights are helpful to ensure an overnight is at the appropriate OD when the lab period begins. Higher ODs will still likely transform but will be less efficient.

Strains that work well for transformation include but are not limited to:

- BY4741 or BY4743 strains classically used in yeast genetic laboratories
- Brewing Strains available at homebrew stores and online. (We have found Safbrew T-58, Lallemend Munich wheat, Lallemend Nottingham Ale, and Lallemend BRY-97 work well.)
- Baking Strains available at grocery stores and online. (We have found Red Star Yeast work well.)

(Multiple transformations with a variety of strains can be performed to compare transformation and gene editing rates between strains.)

2) Pellet 5 OD of cells per transformation by spinning 5 minutes at 5000 RPM in a bench top centrifuge in a 1.5 ml tube. *(For example, if you have 1ml of yeast at OD₆₀₀ = 5 this is equal to 5 total OD and would be enough for one transformation. If your overnight culture is at OD₆₀₀ = 2.5 you will need to pellet 2 mls of cells to have 5 total ODs.)*

- 3) After spinning, use a pipette to remove the supernatant - be careful not to disturb the cells at the bottom of the tube!
- 4) Suspend the pelleted cells in 100µl TE/LiOAc buffer by pipetting up and down until the cells go into solution.
- 5) Add to a 1.5 ml tube the following in the order listed:

Tube 1:

- 1) 100µl cells (from step 4 above)
- 2) 40µl salmon sperm DNA (boiled and cooled on ice)
- 3) More than 10µg pVG1 DNA
- 4) 1ml PLATE buffer

Tube 2:

- 1) 100µl cells
- 2) 40µl salmon sperm DNA (boiled and cooled on ice)
- 3) 0.5µg of pV1382
- 4) 1ml PLATE buffer

Tube 3:

- 1) 100µl cells
- 2) 40µl salmon sperm DNA (boiled and cooled on ice)
- 3) H₂O of the same volume of DNA added to Tube 1 (above)
- 4) 1ml PLATE buffer

- 5) Mix gently by pipetting up and down 10 times.
- 6) Incubate for 30 minutes at 25°C or room temperature.
- 7) Incubate at 42°C for 15 minutes, in either a water bath or heat block. *(This heat shock allows plasmid DNA to get into the cells by creating small openings in the cell membrane.) Alternatively electroporation can be used to perform transformation if equipment is available.*
- 8) Spin 1 minute at max speed in a bench top centrifuge. *(Cells will pellet to the side of the tube and look like a clear film, take care not to disturb these cells when removing the supernatant, using a pipette as done earlier.)*
- 9) Remove PLATE mixture (supernatant) and wash once by adding 1ml of YPD and centrifuging again for 5 minutes at 5000 RPM (~3000x g).

- 10) Suspend in 100µL YPD, allow cells to recover anywhere from 10 minutes to overnight. The longer the recovery the more transformants you will have.
- 11) Plate each transformation on a separate petri dish containing appropriate media by pipetting onto the petri plate and spreading using a cell spreader or glass beads, as directed by your instructor. If using ura- strains such as BY4741, plate onto media lacking uracil Synthetic Complete –Uracil (SC -Ura) if using a prototrophic strain such as a brewing or baking strain plate onto YPD containing 100 µg/ml nourseothricin.
- 12) Colonies will appear and accumulate red color over the next 2-7 days and students should count colonies and record their results in data table 1 during this time period or during the next class.

Data Table 1

Strain	+/- plasmid	# White Colony	# Red Colony

Post Lab Questions:

1. What are the three components of the CRISPR genome editing system and how does each of them contribute to gene editing?
2. Why were the yeast heat shocked during transformation? If we left out the heat shock step, what would be the impact on the number of transformants observed? If we heat shocked at a higher temperature do you think the transformation would work? Why?
3. Why do the *S. cerevisiae* colonies turn red? Why does the red color become more intense over time?
4. Did any *S. cerevisiae* colonies stay white? What would likely be the genotype of a white colony at the *ADE2* locus?

5. What phenotype differences do you see between yeast transformed with pV1382 and pVG1? What part(s) of the plasmids (depicted on Page 5) are responsible for this difference?

6. We transformed 20x more DNA for pVG1 than pV1382. Calculate the number of transformants/ μ g DNA for each plasmid. Which plasmid gave more transformants, corrected for the amount of DNA used? What do you think is responsible for the difference?

7. If we were to mutate a *S. cerevisiae* gene that is not in the adenine biosynthesis pathway, would you expect mutants to be red? If not, how could you determine if you correctly mutated your intended gene?

Thought Questions:

1. Cas9-mediated DNA cleavages can be repaired in two principle ways: 1) Homologous recombination, which uses a repair template 2) Nonhomologous end joining, which does not use a repair template, and relies upon resection of the ends of the DNA followed by ligation of these resected ends. How can you determine which method of repair is used in any individual transformant/colony?

2. You have identified a fungus with a diploid genome and discover a variation in sequence between the alleles of the *ADE2* gene where your gRNA would bind. Describe two ways you could alter your experiment such that you could still mutate both copies of *ADE2*.
3. CRISPR is becoming more and more widely used in a variety of organisms including mammals. Describe one potential benefit and one potential hazard/risk that could come about through genetic engineering. Briefly discuss steps the scientific community could take to ensure that they can realize these benefits while limiting the hazards.
4. In 4-5 sentences summarize the article you found in the news that refers to CRISPR. Provide the URL or cite for your article.

Tips for improving transformation efficiency:

Use more plasmid: Maxi and mega preps are excellent, cost-effective ways to generate large amounts of plasmid.

Allow cells to recover after heat shock for longer. You can let the cells recover overnight before plating if your class schedule allows.

Electroporation is often more efficient than heat shock. Protocol can be found in Vyas et al. (2018).

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Appendix 2: Media Recipes

YPD 1X:

Bacto-Peptone	60 grams
Yeast Extract	30 grams
L-Tryptophan	0.45 grams
D-Glucose (Dextrose)	60 grams
Water	to 3 liters

Dissolve all ingredients by stirring with a stir-bar. Moderate heat may be applied. Aliquot to desired volumes and autoclave 25 minutes.

YEP 2X:

Bacto-Peptone	60 grams
Yeast Extract	30 grams
L-Tryptophan	0.45 grams
Water	to 3 liters

Dissolve all ingredients by stirring with a stir-bar. Moderate heat may be applied. Aliquot to desired volumes and autoclave 25 minutes.

4% Agar

Agar	12 grams
Water	to 300 ml

Autoclave 25 minutes, agar will not dissolve until autoclaved.

YPD Plates with *Nourseothricin*

4% Agar	300 ml
2X YEP	300 ml
40% Weight to Volume Glucose (Dextrose)	30 ml
<i>Nourseothricin to final concentration</i>	100µg/ml

2X Synthetic Complete –Uracil (SC -Ura)

SC drop out Mixture	3.84 grams
Ammonium Sulfate	10 grams
Yeast Nitrogen Base	3 grams
Water	to 1 liter

Dissolve all ingredients by stirring with a stir-bar. Moderate heat may be applied. Aliquot to desired volumes and autoclave 25 minutes.

If red color is not observed after plating on synthetic complete medial alternative SC drop out mixtures with lower amounts of Adenine are commercially available.

Plate Pouring Guidelines:

1. Heat the agar to a boil.
2. Place boiling agar in 65°C water bath and let it cool.
3. Assemble all of the other components for the media in a separate flask and heat to 65°C in a water bath.
4. Combine the agar and the other components of the media mix them by swirling briefly. If you add cold media to boiling agarose you will make a mess and you may burn yourself from boiling over media.
5. Place the mixture back in the water bath at 65°C.
6. If media starts to solidify you can microwave again at this point to dissolve.
7. Once media has cool such that you can touch it with latex gloves for 10 seconds without burning yourself you can add antibiotics. Adding antibiotics to media that is boiling will denature the antibiotics.
8. Label plates with color code on side, I usually also label roughly five plates with the type of plate they are and the date on the lid.
9. Plate media into petri plates (roughly 20 ml per plate) If you do this when the media is too hot it will melt the plates and they will stick together.
10. Let cool on the bench top until solid.
11. YPD they should be left overnight or two nights on the benchtop to dry before being used or put away at 4°C.

Reagents	Catalog Number	Vendor
Yeast Extract, BD Bacto	212750	VWR
Peptone, BD Bacto	211677	VWR
SC-URA drop out media	Y1501-20G	Sigma Aldrich
<i>Nourseothricin</i>	74667-10MG	Sigma Aldrich
Yeast Nitrogen Base (YNB)	Y0626	Sigma Aldrich
Salmon Testes DNA	D7656	Sigma

Appendix 3: Answer Key

Pre Lab Answers:

1. What organism will we be mutating in this lab?

S. cerevisiae, also known as Baker's yeast.

2. What gene will we be mutating in this lab, and how will we be mutating it?

We will be inserting stop codons into *ADE2*.

3. How will we distinguish organisms that have a mutation in the target gene vs those who do not?

Mutated organisms will be red and nonedited will be white.

4. In the sequence in Figure 2, find four additional guide+PAM sequences that are not already labeled:

<u>Guide Sequence (5'→3')</u>	<u>PAM sequence</u>
1. CTGTTGGTATTTAAATTAGG	TGG
2. TTAAATTAGGTGGCCAATT	AGG
3. TAAAATACCACAGTTTGTGTA	TGG
4. GTGCCGGCTTCAACAATCAT	AGG

5. Why must the repair template destroy the recognition sequence for genome editing to occur? What are the two types of recognition sequences that can be changed to accomplish this?

If the recognition sequences are not destroyed CRISPR can come back and cleave the sequence. Guide sequence and PAM sequence.

6. Find and read an article in the news that refers to CRISPR. Be prepared to discuss during the incubation and heat shock portions of the laboratory.

Post Lab Answers:

1. What are the three components of the CRISPR genome editing system and how does each of them contribute to gene editing?

Cas9 – nuclease that cleaves DNA

Guide RNA – guides the nuclease to a specific sequence of DNA

Repair template – repairs the cleaved DNA and introduces the edits to the genome

2. Why were the yeast heat shocked during transformation? If we left out the heat shock step, what would be the impact on the number of transformants observed? If we heat shocked at a higher temperature do you think the transformation would work? Why?

Heat shock increase the permeability of the cell membrane and cell wall allowing the plasmid to enter the cell.

The number of transformants would decrease.

Increasing the heat shock temperature would eventually begin to kill the yeast.

3. Why do the *S. cerevisiae* colonies turn red? Why does the red color become more intense over time?

Buildup of a adenine biosynthesis precursor. Over time more and more of this precursor will accrue.

4. Did any *S. cerevisiae* colonies stay white? What would likely be the genotype of a white colony at the *ADE2* locus?

Wild type or *ADE2*

5. What phenotype differences do you see between yeast transformed with pV1382 and pVG1? What part(s) of the plasmids (depicted on Page 5) are responsible for this difference?

pVG1 red colonies, pV1382 white colonies.

Guide RNA and Repair template responsible for introducing the stop codon in the *ADE2* locus.

- 6. We transformed 20x more DNA for pVG1 than pV1382. Calculate the number of transformants/ μ g DNA for each plasmid. Which plasmid gave more transformants, corrected for the amount of DNA used? What do you think is responsible for the difference?**

In most trials, pVG1 gave less transformants/ μ g than pV1382. While pV1382 does express Cas9 and a guide RNA, the guide does not target the nuclease to cut the yeast genome, while pVG1 has a guide that targets Cas9 to cut at the *ADE2* gene. This cutting of the DNA, is toxic to the cells. Only a subset of the cells properly repair the DNA with the repair template. Instructors should be curve the grading of this to suit the level of the class.

- 7. If we were to mutate a *S. cerevisiae* gene that is not in the adenine biosynthesis pathway, would you expect mutants to be red? If not, how could you determine if you correctly mutated your intended gene?**

No, sequence the DNA of your potential mutants.

Thought Questions:

- 1. Cas9-mediated DNA cleavages can be repaired in two principle ways: 1) Homologous recombination, which uses a repair template 2) Nonhomologous end joining, which does not use a repair template, and relies upon resection of the ends of the DNA followed by ligation of these resected ends. How can you determine which method of repair is used in any individual transformant/colony?**

Sequence the DNA and look to see if your repair template sequence has been incorporated into the genome. If so the yeast have used HR.

- 2. You have identified a fungus with a diploid genome and discover a variation in sequence between the alleles of the *ADE2* gene where your gRNA would bind. Describe two ways you could alter your experiment such that you could still mutate both copies of *ADE2*.**

Sequentially use two different guide RNAs and repair templates that would each target one allele. Alternatively find a region of the gene that is identical between alleles and target it with the same guide sequence.

- 3. CRISPR is becoming more and more widely used in a variety of organisms including mammals. Describe one potential benefit and one potential hazard/risk that could come about through genetic engineering. Briefly discuss steps the scientific community could take to ensure that they can realize these benefits while limiting the hazards.**

Example benefit -- curing disease, making models of disease to test in the lab

Example hazard – unforeseen side effects of making changes to patients DNA causing cancer.

Open dialogue in the scientific community discussing the limits and risks that are considered acceptable in our culture. Potential moratoriums on certain types of editing, Ex germ line editing.

4. In 4-5 sentences summarize the article you found in the news that refers to CRISPR. Provide the URL or cite for your article.